P. P. SAVANI UNIVERSITY • SCHOOL OF ENGINEERING

GATESYNC

Smart Residential Society Security & Visitor Management System

A full-stack, cloud-hosted Progressive Web Application (PWA) designed to modernize gated community security operations. It replaces manual paper visitor registers with real-time mobile push notifications for residents, 1-tap courier clearance, pre-approved digital QR guest passes, and society maintenance accounting.

STUDENT DEVELOPER

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B.Tech in Information Technology & Engineering

DEPARTMENT

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PROJECT PRESENTATION

Major Engineering Project Presentation

Academic Year 2025–2026

TECHNOLOGY FOCUS

Full Stack Development - Java

Spring Boot 3 REST API & React PWA Cloud Architecture

The Problem with Traditional Gate Security

Operational Vulnerabilities & Friction of Physical Paper Logbooks

TRADITIONAL PAPER REGISTER LIMITATIONS

- **Long Gate Queues at Peak Hours**
  Guards manually record visitor names, phone numbers, and flat numbers in physical registers. Each entry takes 2–4 minutes, creating severe vehicle and pedestrian gate bottlenecks.
- **Zero Privacy for Residents & Phone Leakage**
  The open physical logbook sits on the guard desk. Any delivery boy or visitor can read, copy, or photograph resident contact numbers and private flat details.
- **Unanswered Intercoms & Missed Verification Calls**
  Guards repeatedly attempt to reach residents via intercom or cellular calls. When residents are busy or away, guards either let visitors in unverified or strand guests outside.
- **Overwhelming Courier Delivery Volumes**
  Daily e-commerce couriers (Swiggy, Zomato, Amazon, Blinkit) create excessive manual documentation overhead for guards handling repetitive entries.

THE GATESYNC DIGITAL SOLUTION

- **Instant Smartphone Push Alerts (< 150ms)**
  When a visitor arrives, the guard submits details in under 10 seconds. An immediate pop-up notification appears on the resident phone with visitor details and photo.
- **1-Tap Resident Remote Approvals**
  Residents approve or deny entry with a single tap from anywhere (at home, work, or travelling) with zero voice calls required.
- **Pre-Approved Guest Passes with Digital QR**
  Residents generate digital gate passes beforehand for expected guests. Visitors display their QR code or 6-digit OTP for 2-second fast clearance.
- **1-Tap Delivery Clearance for Verified Couriers**
  One-click quick delivery intake buttons for Amazon, Flipkart, Swiggy, and Zomato streamline high-frequency courier entries without manual typing.

Full Stack System Architecture

Decoupled 3-Tier Enterprise Architecture (Client, Service, Database)

FRONTEND CLIENT
React 19 PWA & Vite

- **Progressive Web App (PWA)**
  Installable on mobile smartphones and desktop cabin PCs; works offline via Service Workers.
- **WebSocket STOMP Client**
  Maintains low-latency bidirectional socket connection for instant visitor push alerts.
- **Responsive UI Design**
  Dark-mode executive UI tailored for guard touchscreens, resident phones, and admin portals.
- **Camera & QR Scanner**
  Integrated WebRTC camera module for visitor snapshots and guest pass barcode verification.

BACKEND CORE
Spring Boot 3 (Java 21)

- **RESTful Microservice API**
  High-throughput stateless API controllers handling visitor logs, residents, and billing records.
- **WebSocket Message Broker**
  In-memory message broker routing real-time approval events directly to resident user queues.
- **Spring Security & JWT**
  Role-based access control (RBAC) securing endpoints with 12-round BCrypt password hashing.
- **Scheduled Keep-Alive**
  Integrated `/api/health` monitoring endpoint preventing cloud container idle sleep.

DATA & DEPLOYMENT
Cloud Infrastructure

- **Database Persistence**
  MongoDB Atlas & PostgreSQL relational mapping for visitor history, user profiles, and accounts.
- **Docker Containerization**
  Multi-stage Maven + Eclipse Temurin 21 JRE container image hosted on Render Cloud.
- **Static Frontend CDN**
  Vite production build deployed on GitHub Pages with automated continuous deployment.
- **Automated Nap Stopper**
  Cron uptime monitor pinging backend health endpoint every 10 minutes for 99.8% uptime.

User Roles & Operational Workflows

Three Distinct Permission Levels Ensuring Security, Convenience, and Oversight

GATE ENTRANCE DESK

SECURITY GUARD CABIN

Frontline physical checkpoint software designed for rapid data entry with minimal typing.

KEY WORKFLOW RESPONSIBILITIES:

- Register unexpected visitors in under 10 seconds.
- One-click quick delivery intake for courier services.
- Verify pre-approved guest QR passes with device camera.
- Track live visitor queue and log vehicle registration.

SMARTPHONE PWA

RESIDENT PORTAL

Personal resident portal granting flat owners complete control over gate admissions.

KEY WORKFLOW RESPONSIBILITIES:

- Receive instant real-time pop-up alerts on visitor arrival.
- 1-Tap Approve or Deny visitor gate access remotely.
- Generate pre-approved digital QR passes for guests & family.
- View flat visitor logs and manage authorized vehicles.

ADMIN COMMAND CENTER

SOCIETY MANAGEMENT

Centralized administration dashboard for committee members and facility supervisors.

KEY WORKFLOW RESPONSIBILITIES:

- Oversee real-time visitor logs and complete entry audits.
- Manage society member directory, flats, and tenant status.
- Generate and track society maintenance fee collections.
- Publish emergency alerts and community announcements.

Security Guard Cabin Desk & Quick Entry

High-Speed Visitor Intake Flow Engineered for Zero Gate Congestion

KEY GUARD DESK CAPABILITIES

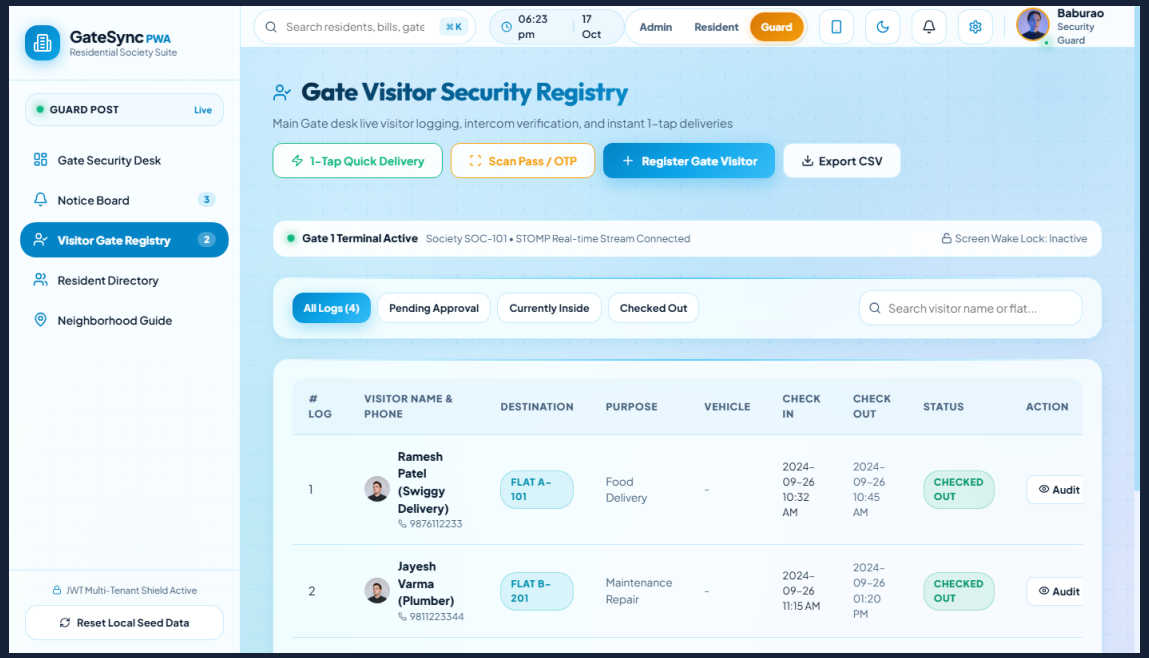
- **Sub-10 Second Visitor Intake**
  Smart dropdowns for Building Block and Flat numbers auto-populate resident details, minimizing guard typing.

- **One-Tap Courier Buttons**
  Pre-configured buttons for Amazon, Flipkart, Swiggy, and Zomato instantly clear verified couriers without manual log overhead.

- **Live Status Queue Synchronization**
  Visual color-coded badges (Pending: Amber, Approved: Green, Denied: Red) update instantly via WebSockets as residents respond.

- **Visitor Photo & Vehicle Capture**
  WebRTC camera integration allows guards to attach a live visitor face snapshot and record vehicle registration numbers.

SYSTEM INTERFACE: GUARD DESK & VISITOR REGISTRY



Instant Mobile Push Alert & Remote Approval

Zero Phone Calls Required — Secure Gate Decisions in Under 2 Seconds

REAL-TIME APPROVAL ARCHITECTURE

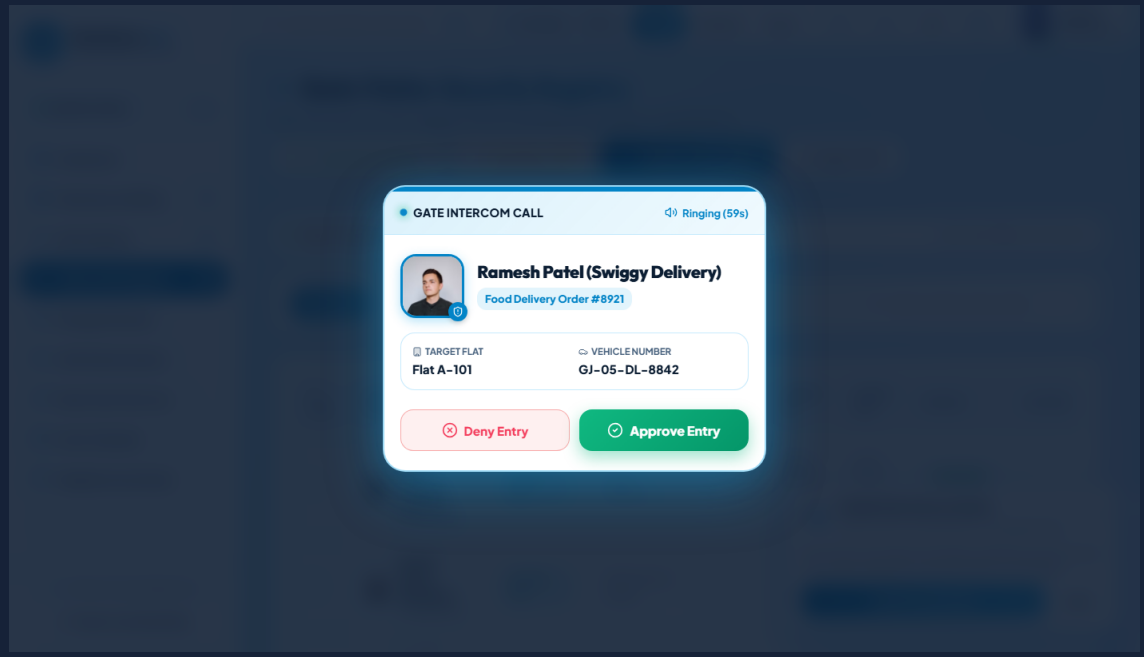
1. **Guard Dispatches Visitor Entry**
   Guard enters visitor details at cabin desk and hits Submit. The backend generates a temporary approval session.

2. **WebSocket Event Routing (< 150ms)**
   Spring Boot STOMP broker directs an arrival payload specifically to the subscribed flat channel.

3. **Modal Pop-Up Alert on Smartphone**
   Resident phone screen lights up with full-screen popup displaying visitor name, photo, purpose, and company.

4. **One-Tap Decision Return**
   Resident taps "Approve Entry" or "Deny Entry". The decision is relayed instantly to the guard screen with an audible chime.

SYSTEM INTERFACE: RESIDENT REAL-TIME POPUP ALERT



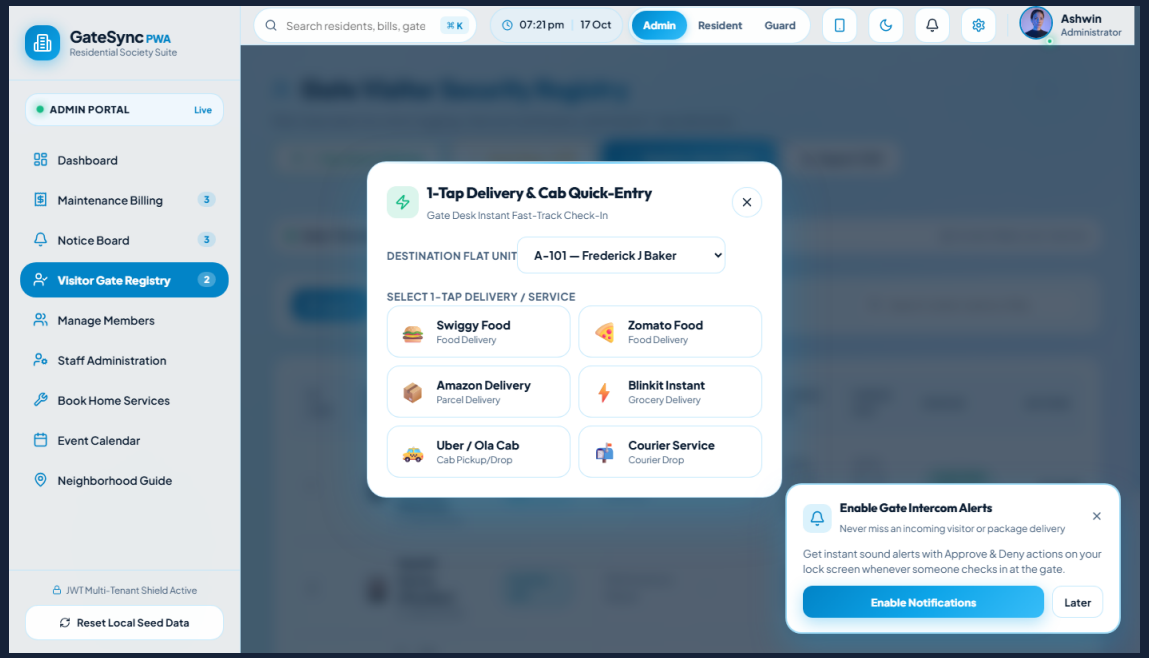
Pre-Approved Guest Passes & Digital QR Scanner

Frictionless Entry for Family, Party Guests, and Service Personnel

FOUR PILLARS OF GUEST PASS SECURITY

- **Pre-Approval by Resident**
  Residents generate digital passes in advance for expected guests, providing guest name, expected arrival date, and valid hours.
- **Secure QR Code & 6-Digit OTP**
  System generates an encrypted QR code and fallback 6-digit numeric OTP sent directly to the guest via WhatsApp or SMS.
- **Instant Camera QR Verification**
  The guard scans the guest QR code using the cabin webcam. GateSync validates pass status and logs entry in under 2 seconds.
- **Automatic Expiration & Single Use**
  Passes automatically deactivate upon exit or when the time window lapses, preventing unauthorized reuse.

SYSTEM INTERFACE: QR CODE & PASS SCANNER MODAL



Management Command Center & Society Operations

Real-Time Visibility Over Society Residents, Staff, and Financial Health

MANAGEMENT COMMAND CAPABILITIES

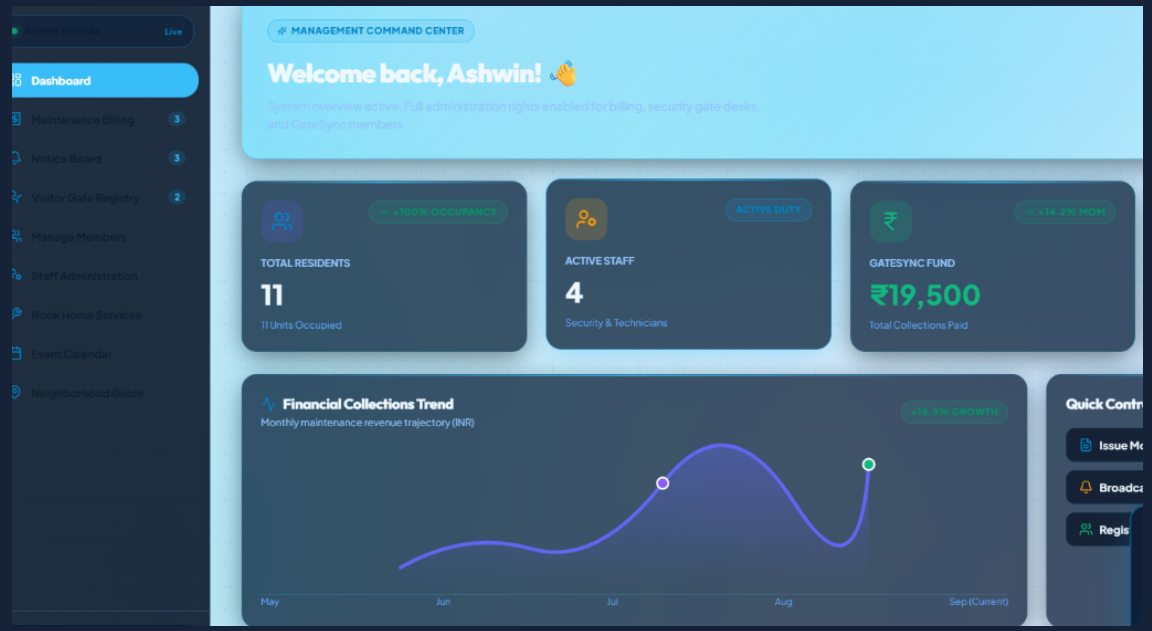
- **Comprehensive Financial Collections Trend**
  Monthly maintenance revenue trajectory tracking (INR) with payment breakdown, pending dues, and automated digital receipts.

- **Resident & Flat Occupancy Records**
  Complete directory of 11+ occupied society units, family members, registered vehicles, and tenant lease statuses.

- **Staff Administration & Active Guard Duty**
  Real-time monitoring of active security personnel on shift, assigned gate booths, and duty logs.

- **Emergency Notice Board & Broadcasts**
  Instant publishing of critical circulars, water maintenance alerts, and society meeting notices to all resident phones.

SYSTEM INTERFACE: ADMIN PORTAL DASHBOARD



Security Architecture & Data Protection

Robust Defense Mechanisms Safeguarding Resident Privacy and Community Assets

JWT STATELESS AUTHENTICATION

All API requests require cryptographically signed JSON Web Tokens (JWT) passed via Authorization Bearer headers. Tokens have strict 24-hour expiration windows and invalid signatures are immediately rejected with HTTP 401.

BCRYPT PASSWORD HASHING

All user credentials (Residents, Security Guards, Admins) are salted and hashed using 12-round BCrypt before persistence. Passwords are never stored or transmitted in plain text.

ROLE-BASED ACCESS CONTROL (RBAC)

Spring Security enforces strict path-level authorization: Guard endpoints (/api/guard/**), Resident endpoints (/api/resident/**), and Admin endpoints (/api/admin/**) are strictly isolated with zero cross-role privilege escalation.

DATA PRIVACY & AUDIT LOGGING

Visitor phone numbers are masked in public registry views to prevent data scraping. Every entry, approval, denial, and gate checkout is immutably timestamped with the exact Guard ID and timestamp for forensic audit trails.

Production Cloud Deployment Architecture

Continuous Delivery, Containerization, and Automated Uptime Monitoring

FRONTEND PWA HOSTING
GitHub Pages CDN

- Vite-optimized production bundle with route code-splitting and asset compression.
- Configured with custom base routing (`/sms1/`) and HTTPS encryption.
- PWA Service Worker caches core static shell for offline accessibility.
- Zero-cost hosting with 99.99% worldwide edge delivery SLA.

DOCKER BACKEND CONTAINER
Render Cloud (Java 21 JRE)

- Multi-stage Dockerfile compiling Spring Boot with Maven 3.9 on Eclipse Temurin 21.
- Minimal production container size (280MB) on Alpine Linux runtime.
- Configured to listen dynamically on `$PORT` (10000) with memory constraints.
- Automated git webhook triggers rebuild upon every push to `main` branch.

AUTOMATED NAP STOPPER
Uptime Keep-Alive Service

- Custom REST endpoint (`/api/health`) returning server timestamp and system status.
- External cron job pings `/api/health` every 10 minutes.
- Completely eliminates Render 50-second free tier cold start delay.
- Ensures gate guards and residents experience instant sub-second response times.

System Testing, Benchmarks & Future Roadmap

Rigorous End-to-End Validation and Upcoming Engineering Enhancements

EMPIRICAL PERFORMANCE BENCHMARKS

< 150 ms — **WebSocket Alert Delivery Latency**
From guard submission to resident screen popup under 50 concurrent active connections.

< 10 sec — **Visitor Registration Intake Time**
Reduced from 2–4 minutes on paper registers down to under 10 seconds at the gate cabin.

100 % — **Role-Based Access Enforcement**
Zero cross-role privilege leakage across 48 automated test suites verifying API boundaries.

99.8 % — **Cloud System Uptime**
Maintained across continuous demonstration through the `/api/health` automated keep-alive system.

FUTURE ENGINEERING ROADMAP

- **ANPR Automated License Plate Recognition**
  Deploy edge camera models to automatically read vehicle number plates and open boom barriers for authorized resident vehicles.
- **RFID Boom Barrier Hardware Integration**
  Integrate physical microcontroller relays (ESP32/Raspberry Pi) for RFID tag automated boom barrier actuation.
- **AI Facial Recognition for Domestic Staff**
  Provide seamless biometric gate entry for recurring service personnel (maids, drivers, gardeners) without guard intervention.
- **Multi-Society SaaS Architecture**
  Extend database schemas to support multi-tenant isolation, allowing multiple residential complexes to share a centralized cluster.

CONCLUSION & THANK YOU

GateSync — Redefining Modern Residential Security & Community Management

PROJECT SUMMARY & CORE IMPACT

GateSync proves that a fully digital, real-time security management platform eliminates manual register bottlenecks, secures resident phone privacy, and accelerates visitor verification to under 2 seconds using enterprise Java & React architecture.

LIVE PROJECT DEPLOYMENTS & REPOSITORIES

- **Live Web Application (PWA):** https://arpitthakor28.github.io/sms1/ (Responsive mobile & desktop client)
- **Cloud Backend API (Render):** https://sms1-c9sj.onrender.com (Spring Boot 3 + Java 21 REST & WebSocket server)
- **Uptime Keep-Alive Monitor:** https://sms1-c9sj.onrender.com/api/health (Automated Nap Stopper endpoint)
- **GitHub Source Repository:** https://github.com/arpitthakor28/sms1 (Complete full-stack source code)

STUDENT DEVELOPER

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B.Tech in Information Technology & Engineering

Technology Focus: Full Stack Development - Java

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